11/04/23

$$(M\omega^2 - K)\,\bar{a} = 0$$

$$\det(M\omega^2 - K) = 0$$

$$\omega_+^2 = 3\omega_0^2$$

$$\omega_-^2 = \omega_0^2$$

$$\bar{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \cos\omega t$$

$$K - M\omega^2 = \begin{bmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 2k - m\omega_+^2 & -k \\ -k & 2k - m\omega_+^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -k & -k \\ -k & -k \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\bar{a} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} a$$

$$\begin{bmatrix} -ka_1 - ka_2 \\ -ka_1 - ka_2 \end{bmatrix} = 0 \Rightarrow a_1 + a_2 = 0$$

$$\boxed{\frac{a_1}{a_2} = -1}$$

$$\Rightarrow \begin{bmatrix} 2k - m\omega_-^2 & -k \\ -k & 2k - m\omega_-^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$ka_1 - ka_2 = 0 \Rightarrow \boxed{a_1/a_2 = 1} \longrightarrow \bar{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} a$$

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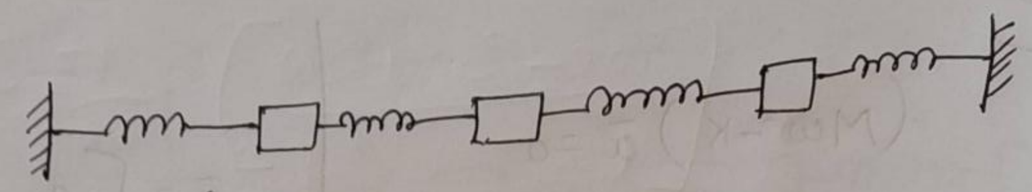

$$K = \begin{bmatrix} 2K & -K & 0 \\ -K & 2K & -K \\ 0 & -K & 2K \end{bmatrix}$$

$$\det(M\omega^2 - K) = 0 \rightarrow \quad \omega_1 = \sqrt{2}\left(1 - \frac{1}{\sqrt{2}}\right)^{1/2}\omega_0, \quad \omega_2 = \sqrt{2}\,\omega_0, \quad \omega_3 = \sqrt{2}\left(1 + \frac{1}{\sqrt{2}}\right)^{1/2}\omega_0$$

$$\begin{bmatrix} 2K - m\omega^2 & -K - m\omega^2 & 0 \\ -K - m\omega^2 & 2K - m\omega^2 & -K - m\omega^2 \\ 0 & -K - m\omega^2 & 2K - m\omega^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

ω_0^2

$\times \frac{A}{K}$

$2K$

$\frac{A}{K}$

$\omega = \omega_1$	$\omega = \omega_2$	$\omega = \omega_3$

$$a = \begin{bmatrix} 1/2 \\ 1/\sqrt{2} \\ 1/2 \end{bmatrix} \quad \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \quad \begin{bmatrix} 1/2 \\ -1/\sqrt{2} \\ 1/2 \end{bmatrix}$$

$\Rightarrow$ Coupled Oscillator Driven:

$$m\ddot{x}_A + m\omega_0^2 x_A + m\omega_c^2 (x_A - x_B) = F_0 \cos\omega t$$

$$m\ddot{x}_B + m\omega_0^2 x_B - m\omega_c^2 (x_A - x_B) = 0$$

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$$\ddot{x}_A + \omega_0^2 x_A + \omega_c^2 (x_A - x_B) = \frac{F_0}{m} \cos \omega t$$

$$\ddot{x}_B + \omega_0^2 x_B + - \omega_c^2 (x_A - x_B) = \cancel{\frac{F_0}{m} \cos \omega t} \; 0$$

<u>adding</u>,

$$(\ddot{x}_A + \ddot{x}_B) + \omega_0^2 (x_A + x_B) = \frac{F_0}{m} \cos \omega t$$

<u>Subs.</u>,

$$(\ddot{x}_A - \ddot{x}_B) + \omega_0^2 (x_A - x_B) + 2\omega_c^2 (x_A - x_B) = \frac{F_0}{m} \cos \omega t$$

$$\left(X_1 = x_A + x_B, \quad X_2 = x_A - x_B \right)$$

$$\boxed{\begin{aligned} &\ddot{X}_1 + \underbrace{\omega_0^2}_{\omega_-} X_1 = \frac{F_0}{m} \cos \omega t \\ &\ddot{X}_2 + \underbrace{(\omega_0^2 + 2\omega_c^2)}_{\omega_+} X_2 = \frac{F_0}{m} \cos \omega t \end{aligned}}$$

$$X_1(t) = A \cos \omega t, \quad A = \frac{F_0/m}{\omega_-^2 - \omega^2}$$

$$X_2(t) = B \cos \omega t, \quad B = \frac{F_0/m}{\omega_+^2 - \omega^2}$$

$$x_A = \frac{X_1 + X_2}{2} = \underbrace{\left(\frac{A+B}{2}\right)}_{C} \cos \omega t$$

$$x_B = \underbrace{\left(\frac{A-B}{2}\right)}_{D} \cos \omega t$$

$$x_A = C \cos \omega t, \quad x_B = D \cos \omega t$$

$$C(\omega) = \frac{A+B}{2} = \frac{F_0}{2m} \left(\frac{1}{\omega_-^2 - \omega^2} + \frac{1}{\omega_+^2 - \omega^2} \right) = \frac{F_0}{2m} \left(\frac{\omega_+^2 - \omega^2 + \omega_-^2 - \omega^2}{(\omega_-^2 - \omega^2)(\omega_+^2 - \omega^2)} \right)$$

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$$C = \frac{F_0}{2m}\left[\frac{\omega_+^2 + \omega_-^2 - 2\omega^2}{(\quad)}\right]$$

$$D = \frac{F_0}{2m}\left[\frac{\omega_+^2 - \omega_-^2}{(\quad)}\right]$$

$$\boxed{\Rightarrow \frac{D}{C} = \frac{\omega_+^2 - \omega_-^2}{\omega_+^2 + \omega_-^2 - 2\omega^2}}$$

$$\Rightarrow \underline{\omega < \omega_-} \qquad 1 > \frac{D}{C} > 0$$

$$\Rightarrow \underline{\omega = 0} \qquad \frac{D}{C} = \frac{\omega_+^2 - \omega_-^2}{\omega_+^2 + \omega_-^2}$$

$$\Rightarrow \underline{\omega = \omega_+} \qquad \frac{D}{C} = -1$$

$$\Rightarrow \underline{\omega = \omega_-} \qquad \frac{D}{C} = +1$$

$$\Rightarrow \underline{\omega > \omega_+} \qquad \frac{D}{C} > -1$$

$$\Rightarrow \underline{\omega \to \infty}$$

$$\boxed{\frac{D}{C} = \frac{\omega_+^2 - \omega_-^2}{-2\omega^2} \qquad -1 < \frac{D}{C} < 0 \qquad \left|\frac{D}{C}\right| < 1}$$

$$\frac{7\omega^2 + \omega^2}{\omega_+^2 - \omega_-^2}$$

$$\frac{(\omega_+^2 - \omega^2) + (\omega_-^2 -}{(\omega_+^2 - \omega^2) - (\omega_-^2}$$

$$\frac{(\omega_+^2 - \omega^2) + (\omega_-^2 -}{(\omega_-^2 - \omega_-^2) - (\omega_-^2 - \omega_+^2)}$$

$$\frac{(\omega^2 - \omega_+^2) + (\omega_-^2}{}$$

$$-1 < \frac{D}{C}$$

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